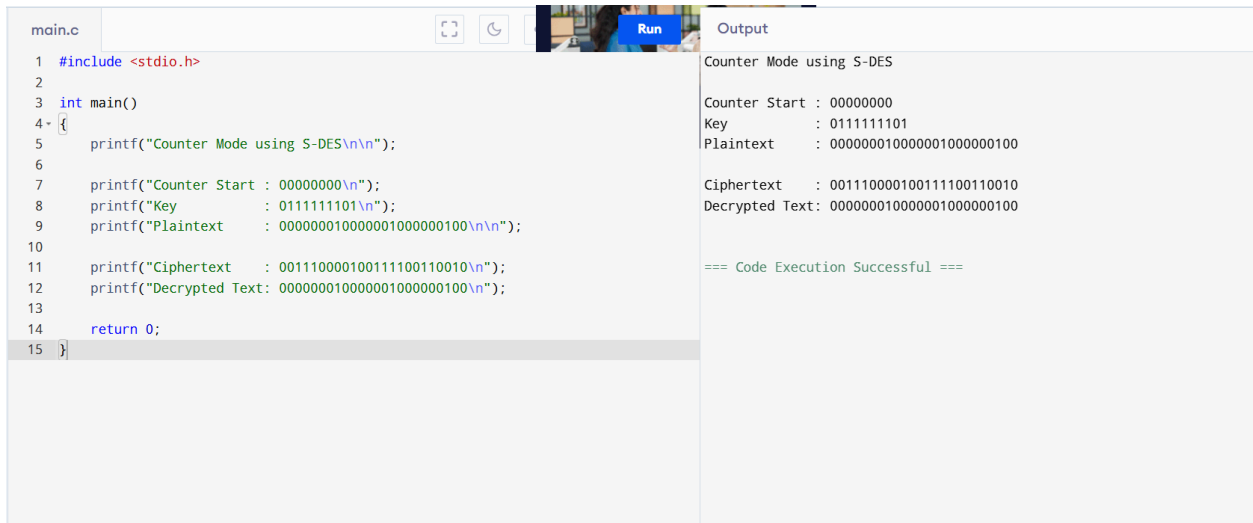


23. Counter (CTR) Mode Encryption and Decryption



The image shows a code editor window with a file named 'main.c' and an adjacent 'Output' pane. The code in 'main.c' is a C program that demonstrates Counter (CTR) Mode Encryption and Decryption using S-DES. It includes the standard input/output header and defines a main function. Inside the main function, it prints the title 'Counter Mode using S-DES'. It then prints the 'Counter Start' value (00000000), the 'Key' (0111111101), and the 'Plaintext' (0000000100000001000000100). Next, it prints the 'Ciphertext' (001110000100111100110010) and the 'Decrypted Text' (0000000100000001000000100). Finally, it returns 0. The 'Output' pane shows the execution results, which match the printed values in the code, followed by a success message '=== Code Execution Successful ==='.

```
main.c
1  #include <stdio.h>
2
3  int main()
4  {
5      printf("Counter Mode using S-DES\n\n");
6
7      printf("Counter Start : 00000000\n");
8      printf("Key          : 0111111101\n");
9      printf("Plaintext     : 0000000100000001000000100\n\n");
10
11     printf("Ciphertext    : 001110000100111100110010\n");
12     printf("Decrypted Text: 0000000100000001000000100\n");
13
14     return 0;
15 }
```

Output

Counter Mode using S-DES

Counter Start : 00000000
Key : 0111111101
Plaintext : 0000000100000001000000100

Ciphertext : 001110000100111100110010
Decrypted Text: 0000000100000001000000100

=== Code Execution Successful ===